REINVENT COMMERCE

ShelfCast

An AI demand planner on an **AMD Instinct MI300X** that turns a forecast into the right order.

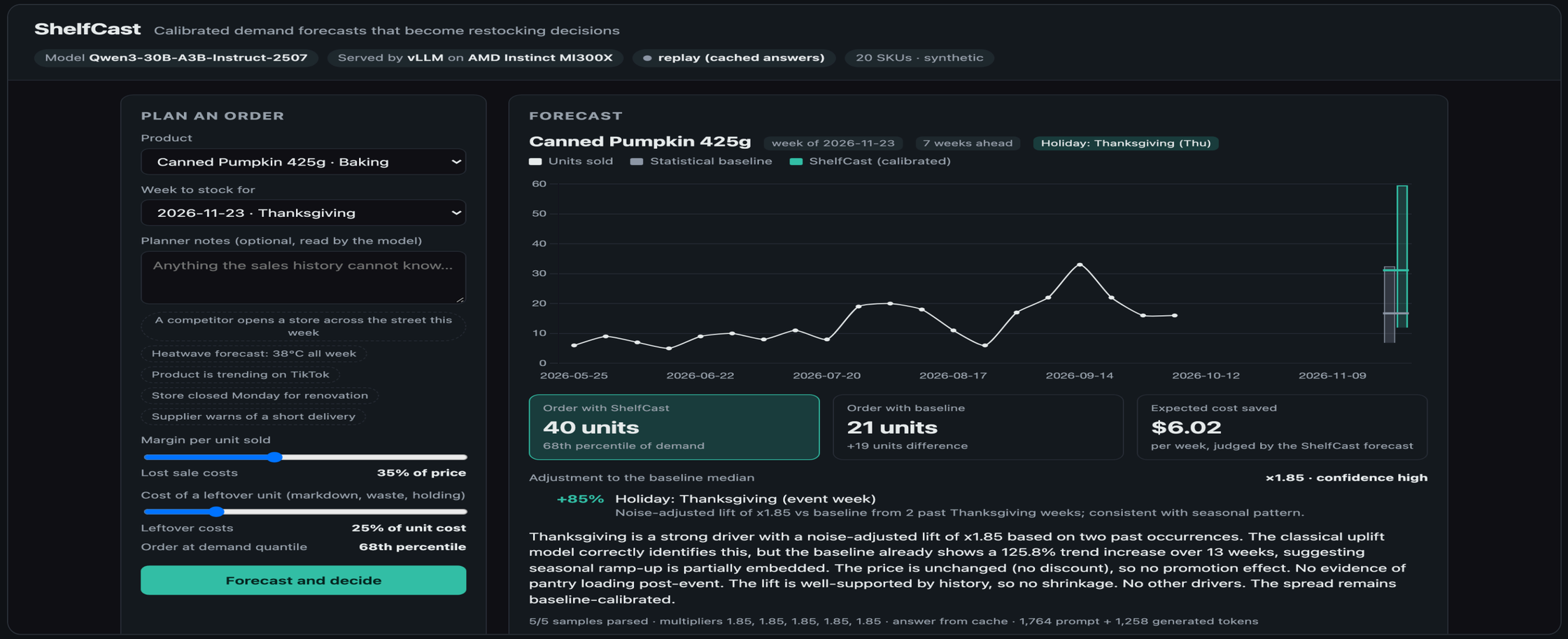
github.com/yy5652-hash/shelfcast
yy5652-hash.github.io/shelfcast

THE PROBLEM

How many should we order?

- Every week, every store, every product. Too few is an empty shelf; too many is waste.
- Replenishment systems extrapolate last month. They cannot see a holiday, a promotion, or what the store manager knows.
- Planners patch the number by hand: slow, inconsistent, and never checked against what sold.

WHAT THE PLANNER SEES



One screen: the range, the order, each driver with its evidence, and the exact brief the model read.

ShelfCast · AMD Developer Hackathon ACT III

HOW IT WORKS

A statistical forecast, then judgment, then a decision.

1

Baseline

Exponential smoothing with quantiles from its own past errors.

2

Evidence

Past event and promotion effects, kept only if they clear the product's own noise.

3 · ON AMD

LLM judgment

Reads the evidence and the planner's notes; returns a multiplier, two spreads, and reasons.

4

Guard + calibration

No evidence and no note: the baseline stands. Ranges recalibrated on past errors.

5

Order

The demand quantile where lost margin balances leftover cost.

The language model is never the only forecasting method.

BUILT ON AMD

One AMD Instinct MI300X. Open weights. vLLM on ROCm.

256

forecasts a minute

128 concurrent, 5 samples each

4,722

output tokens a second

p95 latency 47.8s

$0.19

per 1,000 forecasts

at $2.99 per GPU-hour

Model **Qwen/Qwen3-30B-A3B-Instruct-2507**. Each brief is prefilled once for all samples; guided decoding enforces the JSON. Sales data never leaves the retailer's GPU.

DOES IT PAY? REAL WALMART SALES (M5)

1,560 forecasts, each made only from the past.

-8.0%

inventory cost

ordering the cost-optimal quantile instead of the point forecast

-0.6%

forecast accuracy vs baseline

pinball loss, positive is better; event weeks -1.7%

90%

of outcomes inside the 90% range

target 90%

60 item-store series, 2015-11-21 to 2016-05-14, model Qwen/Qwen3-30B-A3B-Instruct-2507. These items are anonymised and carry no promotion data, so there is little for judgment to add: the gain here is the decision layer.

WHERE JUDGMENT EARNS ITS PLACE

When there is something to read, the model reads it.

-15.7%

inventory cost in holiday and promotion weeks

synthetic store, vs calibrated baseline

+19.4%

forecast accuracy in weeks with a planner note

pinball loss vs calibrated baseline; positive is better

+17.6%

the same, vs a rule with no LLM

the baseline and the rule cannot read a note

A generated store with holidays, promotions and free-text notes such as "supplier warns this week's delivery will be short" (no number given). Labelled synthetic throughout.

WHAT WE LEARNED, AND WHAT IS NEXT

We ran the ablations before the pitch.

- Left free, the model made forecasts **worse** on real data: it read routine calendar entries as events. So it is the judgment layer, tied to evidence by a guard.
- Its own contribution is measured apart from everything else, against a rule that uses the same evidence.
- Limits: one week ahead, no carry-over stock, assumed margins.
- Next: multi-week horizons, real promotion calendars, planner feedback as training data.

SHELFCAST

From forecast to the right order.

Open model. Your GPU. Your data stays home.

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